

tf.compat.v1.variance_scaling_initializer

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/variance_scaling_initializer

Initializer capable of adapting its scale to the shape of weights tensors.

Function Signatures

```
tf.compat.v1.variance_scaling_initializer( scale=1.0,  
mode='fan_in', distribution='truncated_normal', seed=None,  
dtype=tf.dtypes.float32 )
```

Code Examples

```
tf.compat.v1.variance_scaling_initializer(  
scale=1.0,  
mode='fan_in',  
distribution='truncated_normal',  
seed=None,  
dtype=tf.dtypes.float32  
)
```

```
tf.compat.v1.variance_scaling_initializer(  
    scale=1.0,  
    mode='fan_in',  
    distribution='truncated_normal',  
    seed=None,  
    dtype=tf.dtypes.float32  
)
```

```
tf.dtypes.float32
```

```
initializer = tf.compat.v1.variance_scaling_initializer(  
    scale=scale,  
    mode=mode,  
    distribution=distribution  
    seed=seed,  
    dtype=dtype)  
weight_one = tf.Variable(initializer(shape_one))  
weight_two = tf.Variable(initializer(shape_two))
```

```
initializer = tf.compat.v1.variance_scaling_initializer(  
    scale=scale,  
    mode=mode,  
    distribution=distribution  
    seed=seed,  
    dtype=dtype)  
weight_one = tf.Variable(initializer(shape_one))  
weight_two = tf.Variable(initializer(shape_two))
```

```
initializer = tf.keras.initializers.VarianceScaling(  
    scale=scale,  
    mode=mode,  
    distribution=distribution  
    seed=seed)  
weight_one = tf.Variable(initializer(shape_one, dtype=dtype))  
weight_two = tf.Variable(initializer(shape_two, dtype=dtype))
```

```
initializer = tf.keras.initializers.VarianceScaling(
    scale=scale,
    mode=mode,
    distribution=distribution
    seed=seed)
weight_one = tf.Variable(initializer(shape_one, dtype=dtype))
weight_two = tf.Variable(initializer(shape_two, dtype=dtype))
```

```
@classmethod
from_config(
    config
)
```

Documentation

Initializer capable of adapting its scale to the shape of weights tensors.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.initializers.variance_scaling`

Migrate to TF2

Although it is a legacy `compat.v1` API, this symbol is compatible with eager execution and `tf.function`.

To switch to TF2 APIs, move to using either

`tf.initializers.variance_scaling` or `tf.keras.initializers.VarianceScaling`
(neither from `compat.v1`) and
pass the `dtype` when calling the initializer.

Structural Mapping to TF2

Before:

After:

How to Map Arguments

Description

With `distribution="truncated_normal"` or `"untruncated_normal"`,
samples are drawn from a truncated/untruncated normal
distribution with a mean of zero and a standard deviation (after truncation,
if used) $\text{stddev} = \sqrt{\text{scale} / n}$
where `n` is:

- number of input units in the weight tensor, if `mode = "fan_in"`
- number of output units, if `mode = "fan_out"`
- average of the numbers of input and output units, if `mode = "fan_avg"`

With `distribution="uniform"`, samples are drawn from a uniform distribution
within `[-limit, limit]`, with $\text{limit} = \sqrt{3 * \text{scale} / n}$.

Raises

Methods

from_config

[View source](#)

Instantiates an initializer from a configuration dictionary.

Example:

```
## get_config
```

[View source](#)

Returns the configuration of the initializer as a JSON-serializable dict.

__call__

[View source](#)

Returns a tensor object initialized as specified by the initializer.